```
main(){
    char v='2';
    printf("%c %d\n", v, v);
    v=v*2;
    printf("%c %d\n", v, v);
    v=v*2;
    printf("%c %d\n", v, v);
    v=v*2;
    printf("%c %d\n", v, v);
}
```

v → 50 (100)

v is 1 byte (8 bits)

1 bit [] => 0/1 => 2

[][] => => 4

8 bits: $2^8$ => 256

signed char x;

unsigned char y;

char z; // signed by default

x: 7 6 5 4 3 2 1 0

y: 7 6 5 4 3 2 1 0

2 50

-56

-112

undefined

<typemodifier> var_Decl;

signed

unsigned

long

short

Range of char - 1 byte - 8 bits - $2^8$ = 256

256

0 ... 255

-128 ... -1, 0 ... 127

20

128 64 32 16 8 4 2 1

0 0 0 1 0 1 0 0

2 | 20

2 | 10 - 0

2 | 5 - 0

2 | 2 - 1

1 - 0

main(){

char ch = -50;

Represent ch in memory - 8 bits.

-ve Numbers Are stored in 2's Compliment form

char x = 200;  200 → ?

(%d, x); // -56

128 128+64+8

| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 |

x  -56

00 11 0 1 1 1

+ 1

00 11 1000

32 16 8

short int v1;

unsigned short int v2;

signed short int v3;

%hd %hu

2 bytes → 16bits ⇒ $2^{16}$ ⇒ $2^{10} * 2^6$

1K x 64 ⇒ 64K

65536

7bit ASCII

0 ... 127

Range

- signed : -32768 to 32767
- unsigned : 0 ... 65535

int 4byte / 2byte

int x;

unsigned int y;

} 4bytes ⇒ 32bits ⇒ $2^{32}$ ⇒ $2^{10} \times 2^{10} \times 2^{10} \times 2^2$

1K X1K

1M X1K

1Gi X 4

4Gi

%d

%u/%o/%x

65536 x 65536

4,000,000,000

295 ...

4.3Gi

long int : 4 byte or 8byte

%ld %lu

↳ 64 bits

↳ $2^{64}$

$2^{64} = 2^{32} \times 2^{32} = 4.3 \times 10^9 \times 4.3 \times 10^9$

$= 18.2 \times 10^{18}$

$= 1.8 \times 10^{19}$

---

long long int : 8 bytes

↳ %lld %llu

Array or Struct

50 => 00110010

1's (50) => 11001101

+ 1

2's n => 11001110 (-50)

---

Char ch = -1;

What's the Bin. eq. of ch.?

1 => 00000001

1's (1) => 1111 1110

+ 1

1111 1111 -1

---

```
main() {
    char c1 = 127, c2;
    c2 = c1+1;
    Pf("%d %d", c1, c2); // 127, -128
}
```

c1 => 0111 1111 => 127

c2 = 128 | 1000 0000 | -n (-128)

S 7

In a signed char variable & char x;

The bin found as | 11110100 | S -n = (-12)

What is the Decimal equivalent?

-n is 2's comp of n

n is 2's comp of -n

0000 1011

+ 1

0000 1100 n = 12

var = 12345 ⟨ %f ⇒ 12345.500000
⟨ %e ⇒ 1.234500 e+3

± D.ddddd e ± n
[1-9]

↳ 1.234500 e+6
1234500.0---

~~0.000567~~8
5.678 e-4

123.4
↳ 123.400001

123.3
↳ 123.29999

⇒ Some of the floating point data are slightly truncated.

⇒ The range of float (4bytes) is much higher than $2^{32}$.

→ Unsigned float not allowed.

---

IEEE-754